

Summarizing Data (Solutions)

View the 'big picture'

Before doing any analysis, always explore your data.

What does the data look like?

Is the data appropriate for my chosen analysis?

How can I use it to answer my research question?

Use both **visual** and **numeric** summaries.



The Palmer penguins dataset records physical measurements of penguins taken at Palmer Research Station.

Each row is a different penguin individual.

- Three species
- Measured at different locations/times
- Several physical characteristics + sex

Exercises

How can we use this data to learn about the physical differences between the three species?

- Discuss ideas for a *visualization*
- Discuss ideas for a *numeric* summary
- What information is *lost* when we make these summaries?

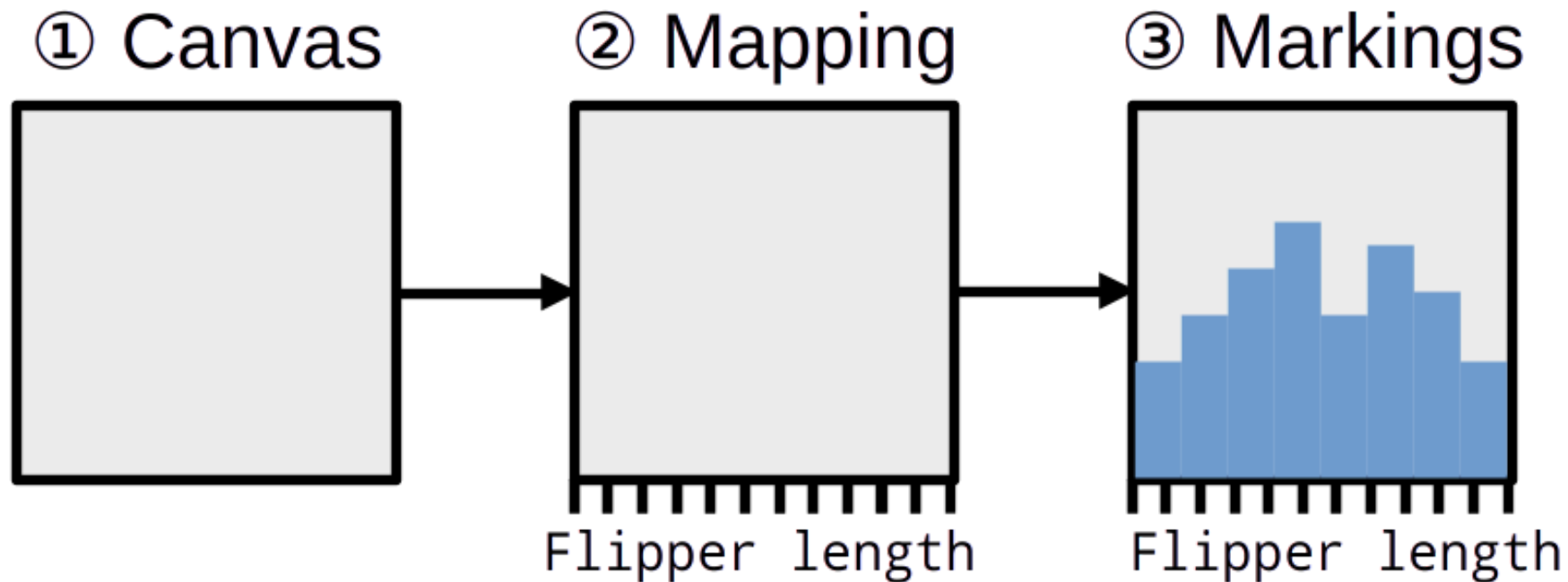


Solutions/Hints

There is no one right answer here. Consider flipper length - what does the *average* flipper length tell us, as opposed to the *median* flipper length?

If you aren't sure, don't worry! We'll go through explicit definitions later.

The `tidyverse` package includes a “grammar of graphics” `ggplot()`.



Single numeric variable

Histograms (`geom_histogram`), box plots (`geom_boxplot`)

Single categorical variable

Bar graph (`geom_bar`), column graph (`geom_col`)

Two numeric variables

Scatterplot (`geom_point`), line plot (`geom_line`)

`ggplot2` offers rich customization options for:

- Color
- Shape
- Size
- Transparency

and more...

These don't change the *information* shown in the graph.

Exercises

Choose one of the penguin data graphs we made and change the style from the default output.

- Change the color, size or transparency of at least one visual element.
- Look up the help file for the `geom_` function and figure out which aesthetic options are available.
- CHALLENGE: How is `color` different from `fill`?



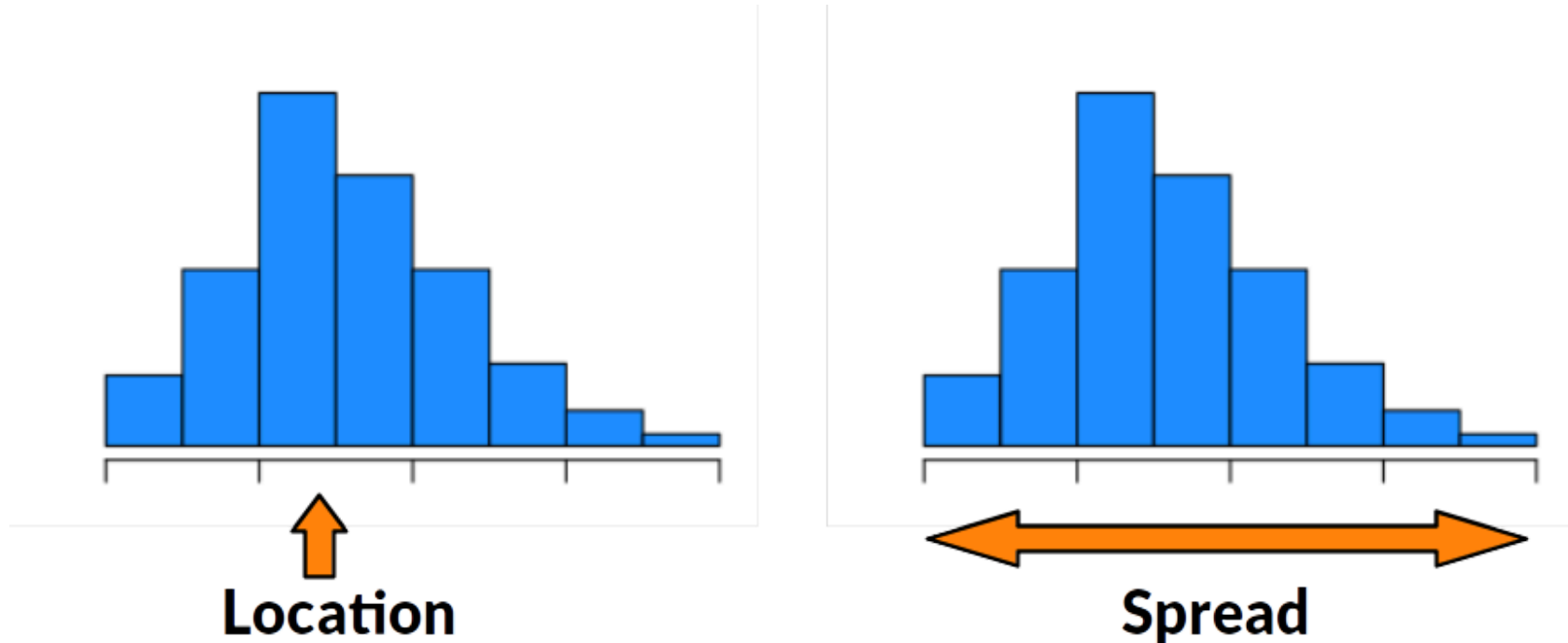
Solutions/Hints

If you're just changing the style, make sure the options go inside of the `geom()`, not inside of the original `ggplot()`.

Look up the help file with `?`. For example, run `geom_bar()` in the R console.

If you scroll through the help file you'll see a section called "Aesthetics" showing the different options.

Numeric summaries help quantify our data.



Location: mean, median, quantiles

Spread: sd, range, IQR

$$\text{Mean} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

“typical value”

$$\text{Standard deviation} = s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

“typical deviation from mean”

The **median** is the midpoint of the data.

This generalizes to **percentiles**, e.g. 30th or 75th percentile.

Percentiles refer to the amount *below*!

SD is based on mean - other measures of spread rely on the percentiles.

The **range** is the “footprint” of the entire dataset.

The **inter-quartile range (IQR)** is the “footprint” of the middle 50% of the data.

Consider the following two statements:

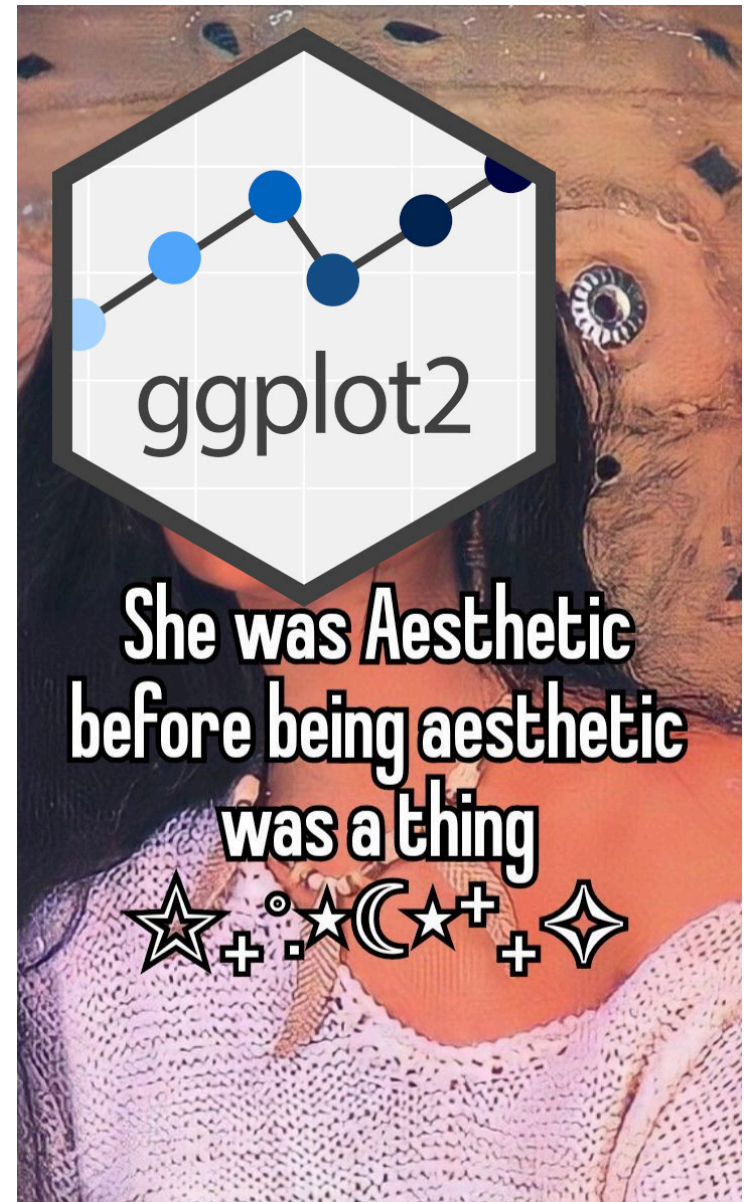
- “The average midterm score is 70/100”
- “Half of the students did worse than 70/100, and half did better”

Are these always both true? How could they be different?

Consider whether summaries are *robust* to outliers.

Let's look closer at ggplot
aesthetics.

Color, fill, x, y, alpha, etc.



variable
↓
ggplot(aes()) +
geom()
↑
constant

Aesthetics in our `aes( )` mapping are **variable aesthetics**.

- Tied to a specific column (variable)

Constant aesthetics change the style of the plot.

- Not mapped to any column

What if we want a mapping to apply to only one layer?

Variable aesthetics can be either **global** or **local**.

(global) variable
↓
ggplot(aes()) +
geom(aes(, )
↑
(local) variable

Exercises

The four `ggplot()` calls return either an error or an unexpected result. Identify and fix the problem with each one.

- Why does `geom_point()` not work?
- Why can't R find the `species` column?.
- Why are the points not huge? Why is there a legend for “size”?
- Why is there a blank gridded canvas?.



Solutions/Hints

- `geom_point()` requires us to specify `x` and `y`.
- The column name `species` must go in `aes()`, which defines the mapping.
- We want `size` to be constant, so it should go outside of `aes()`.
- We need a `+` to add the point layer to the plot!

The `ggplot2` package offers lots of customization tools.

- Text annotations
- Axis scales + labels
- Color palettes

(Download the slide .pdf if the links don't work.)

Looking ahead

In [chapter 5](#), we'll learn how to tidy and clean data. If you want a head start, run the following lines of code to create two different summary tables (make sure you have [tidyverse](#) and the penguin data loaded):

```
1 penguins %>%  
2   summarize(avg_flipper_len = mean(flipper_length_mm))  
3  
4 penguins %>%  
5   group_by(species) %>%  
6   summarize(avg_flipper_len = mean(flipper_length_mm))
```

What is the difference between the two summary tables? What do you think the `%>%` symbol is doing?